

EXPERIMENT 9

MEGASCOPIC STUDY OF ECONOMIC MINERALS

Structure

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9.1 INTRODUCTION

You have studied about physical properties of some of the common rock-forming minerals in the Experiments 4 and 5, and about optical properties in Experiments 7 and 8. The study of physical and optical properties of a mineral is useful to identify that mineral in the field and laboratory respectively. Economic minerals/ores (both metallic and non-metallic) are the basic building blocks of the modern industrialised society. Minerals of the economic importance are the subject matter of the Economic Geology. In this experiment, you will get hands-on experience of identifying metallic and non-metallic minerals/ores in hand specimens. In this experiment, you will work with the physical properties of

some of the common ore minerals (iron, manganese, copper, lead, zinc and aluminium) and metallic minerals (mica, gypsum, magnesite and asbestos).

Expected Skills

After performing this experiment, you should be able to:

- ❖ identify physical properties of iron, manganese, copper, lead, zinc and aluminium ores; and
- ❖ recognise non-metallic minerals/ores such as mica, gypsum, magnesite and asbestos.

9.2 REQUIREMENTS

To perform this experiment successfully, following are the requirements:

- Pen, pencil, eraser, scale, sharpener, coloured pencils and drawing compass
- Laboratory File
- Hand specimens of iron, manganese, copper, lead, zinc and aluminium ores and non-metallic mica, gypsum, magnesite and asbestos minerals.

Note:

- **Do not use pen/pencil/ marker pen to mark the hand specimen of the mineral.**
- **Please do not attempt to cleave the minerals in the laboratory. Many of the specimens you examine cannot be readily replaced.**
- **Do not break the mineral/ores in order to see its cleavage. Cleavage is usually induced in the mineral when it is extracted from the rock it is found in and is usually seen as planes running through the mineral.**

Instructions: You are required to study Units 13 and 14 of BGYCT-133 course (Crystallography, Mineralogy and Economic Geology) before performing this experiment. Bring this practical manual along with Block 4 of BGYCT-134 course while attending the Practical Counselling session.

9.3 BASIC CONCEPTS

You have already studied about the physical properties of minerals and based on them you have identified common rock-forming minerals in Experiments 4 and 5. This study of minerals is also known as **megascopic study** and is the cheapest and fastest method of identification for ores and minerals. Form (habit), colour, cleavage, fracture, hardness, specific gravity, lustre, streak and magnetism are some of the common physical (megascopic) properties of the minerals. The same physical properties are used to identify ores and non-metallic minerals in this experiment.

You will use the following physical properties of minerals to identify a mineral:

1. Form/Habit
2. Colour
3. Cleavage
4. Fracture

5. Hardness
6. Specific Gravity
7. Lustre
8. Streak

Study all the minerals one by one and write the physical properties of the ore/mineral specimens in the manner as listed below in Table 9.1. Also, mention diagnostic physical properties in the laboratory file as used by you in the identification of a particular mineral/ore.

Table 9.1: Format for describing ore/mineral specimens.

S.N.	Physical properties	Characteristics
1.	Chemical composition	
2.	Crystal system	
3.	Form/Habit	
4.	Colour	
5.	Cleavage	
6.	Fracture	
7.	Hardness	
8.	Specific gravity	
9.	Lustre	
10.	Streak	
11.	Special property	
12.	Diagnostic properties	

9.4 PHYSICAL PROPERTIES OF METALLIC ORES

Let us examine the physical properties of ores/minerals of iron, manganese, copper, lead, zinc and aluminium metals.

9.4.1 Iron Ores

Iron is abundantly present on the crust of the Earth and it accounts for more than 95% of all metals used by the human society. Iron is mostly found in rocks in the forms of oxides, carbonates, sulphides and silicates that you have read in unit 13 of BGYCT-133 course. **The common iron ores are hematite, magnetite and pyrite.** You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.2 for hematite (Fig. 9.1a & b); Table 9.3 for magnetite (Fig. 9.2a) and Table 9.4 (Fig. 9.2b); for pyrite.



(a)

(b)

Fig. 9.1: Hematite: a) Massive form; and b) Reniform (kidney ore).

Table 9.2: Physical properties of hematite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	Fe_2O_3 (Fe: 70%)
2.	Crystal system	Hexagonal
3.	Form/Habit	Massive (Fig. 9.1a), tabular, crystalline, oolitic, reniform (Fig. 9.1b), amorphous
4.	Colour	Steel gray to black, bright red
5.	Cleavage	None
6.	Fracture	Uneven to sub-conchoidal
7.	Hardness	5.5 - 6.5
8.	Specific gravity	4.9 - 5.3
9.	Lustre	Metallic, submetallic
10.	Streak	Cherry-red to red-brown
11.	Special property	Weakly magnetic
12.	Diagnostic properties	Forms - Reniform, oolitic; differs from other ores of iron such as magnetite, ilmenite and goethite Streak - Cherry-red Hardness - High Weakly magnetic

Table 9.3: Physical properties of magnetite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	Fe_3O_4 (Fe: 72%)
2.	Crystal system	Cubic/Isometric
3.	Form/Habit	Massive (Fig. 9.2a), granular
4.	Colour	Steel grey to black
5.	Cleavage	None
6.	Fracture	Conchoidal to sub-conchoidal
7.	Hardness	5.5 - 6.5
8.	Specific gravity	5.1
9.	Lustre	Metallic, sub-metallic
10.	Streak	Black
11.	Special property	Weakly magnetic
12.	Diagnostic properties	Streak - black and strongly magnetic

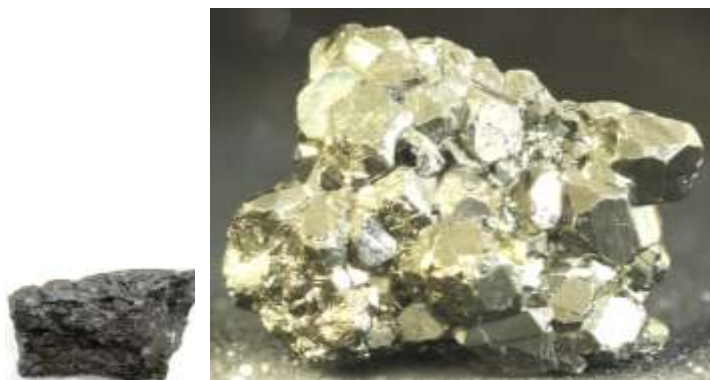


Fig. 9.2: a) Magnetite in massive form; and b) Aggregate of pyrite crystals.

Table 9.4: Physical properties of pyrite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	FeS_2 (Fe: 46.6%)
2.	Crystal system	Cubic/Isometric
3.	Form/Habit	Cubic (Fig. 9.2b), crystalline, massive, nodular, globular, radial-fibrous
4.	Colour	Pale brass yellow, bronze yellow, yellowish brown
5.	Cleavage	Indistinct
6.	Fracture	Uneven, conchoidal
7.	Hardness	6 - 6.5
8.	Specific gravity	4.8 - 5.1
9.	Lustre	Metallic, splendid
10.	Streak	Black
11.	Special property	Weakly magnetic
12.	Diagnostic properties	Colour - Pale brass yellow, bronze yellow Form/Habit – Cubic crystals Hardness - High, Lustre - Metallic and splendid Streak - Black and strongly magnetic

9.4.2 Manganese Ores

Manganese is a transition metal. It occurs in association with iron and in combination with other minerals. The common ores of manganese are oxides that you have read in unit 13 of BGYCT-133 course. The common manganese ores are psilomelane and pyrolusite. You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.5 for psilomelane (Fig. 9.3a) and Table 9.6 for pyrolusite (Fig. 9.3b).

Table 9.5: Physical properties of psilomelane.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$[\text{BaMn}_3(\text{OH})_6\text{Mn}_8\text{O}_{18}]$
2.	Crystal system	Monoclinic
3.	Form/Habit	Massive, botryoidally (Fig. 9.3a) reniform, radial-fibrous
4.	Colour	Iron black
5.	Cleavage	None
6.	Fracture	Brittle
7.	Hardness	5 – 6
8.	Specific gravity	3.7 - 4.7
9.	Lustre	Submetallic
10.	Streak	Brownish black with shine
11.	Diagnostic properties	Colour - Iron black Form - Massive, botryoidal Hardness - 5 – 6 Streak - Brownish black with shine



(a)



(b)

Fig. 9.3: Manganese ores: a) Psilomelane in botryoidal form; and b) Pyrolusite in massive form.

Table 9.6: Physical properties of pyrolusite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	MnO ₂ (Mn: 63%)
2.	Crystal system	Orthorhombic
3.	Form/Habit	Massive (Fig. 9.3b), reniform, fibrous, radiate
4.	Colour	Black, iron gray, dark steel gray
5.	Cleavage	Perfect, poor
6.	Fracture	Brittle (soils finger)
7.	Hardness	2 - 2.5
8.	Specific gravity	4.8 – 5
9.	Lustre	Submetallic
10.	Streak	Black, bluish black
11.	Diagnostic properties	Hardness - Low Streak - Black, bluish black Fracture - Brittle

9.4.3 Copper Ores

Native copper is an element and a mineral. It is rarely found in large quantities. Most copper produced is extracted from sulphide deposits. **The common copper ores are chalcopyrite, malachite and cuprite.** You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.7 for chalcopyrite (Fig. 9.4a), Table 9.8 for malachite (Fig. 9.4b) and Table 9.9 for cuprite (Fig. 9.4c).

Table 9.7: Physical properties of chalcopyrite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	CuFeS_2 (Cu: 34.5%)
2.	Crystal system	Tetragonal
3.	Form/Habit	Commonly massive (Fig. 9.4a)
4.	Colour	Golden yellow
5.	Cleavage	Imperfect, indistinct
6.	Fracture	Conchoidal, uneven, brittle
7.	Hardness	3.5 - 4
8.	Specific gravity	4.1 - 4.3
9.	Lustre	Metallic
10.	Streak	Greenish black
11.	Diagnostic properties	Colour - Golden yellow Hardness – Medium (3.5 – 4) Streak - Greenish black

Table 9.8: Physical properties of malachite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{CuCO}_3\text{Cu}(\text{OH})_2$ (Cu: 57.3%)
2.	Crystal system	Monoclinic
3.	Form/Habit	Massive, botryoidally granular (Fig. 9.4b), earthy, reniform, encrusting, radial, fibrous
4.	Colour	Green, bright green
5.	Cleavage	Perfect, distinct
6.	Fracture	Conchoidal, uneven, brittle
7.	Hardness	3.5 – 4
8.	Specific gravity	3.7 - 4.0
9.	Lustre	Silky, dull, vitreous
10.	Streak	Pale green
11.	Diagnostic properties	Colour - Bright green Form - Radial, fibrous and encrusting Streak - Pale green

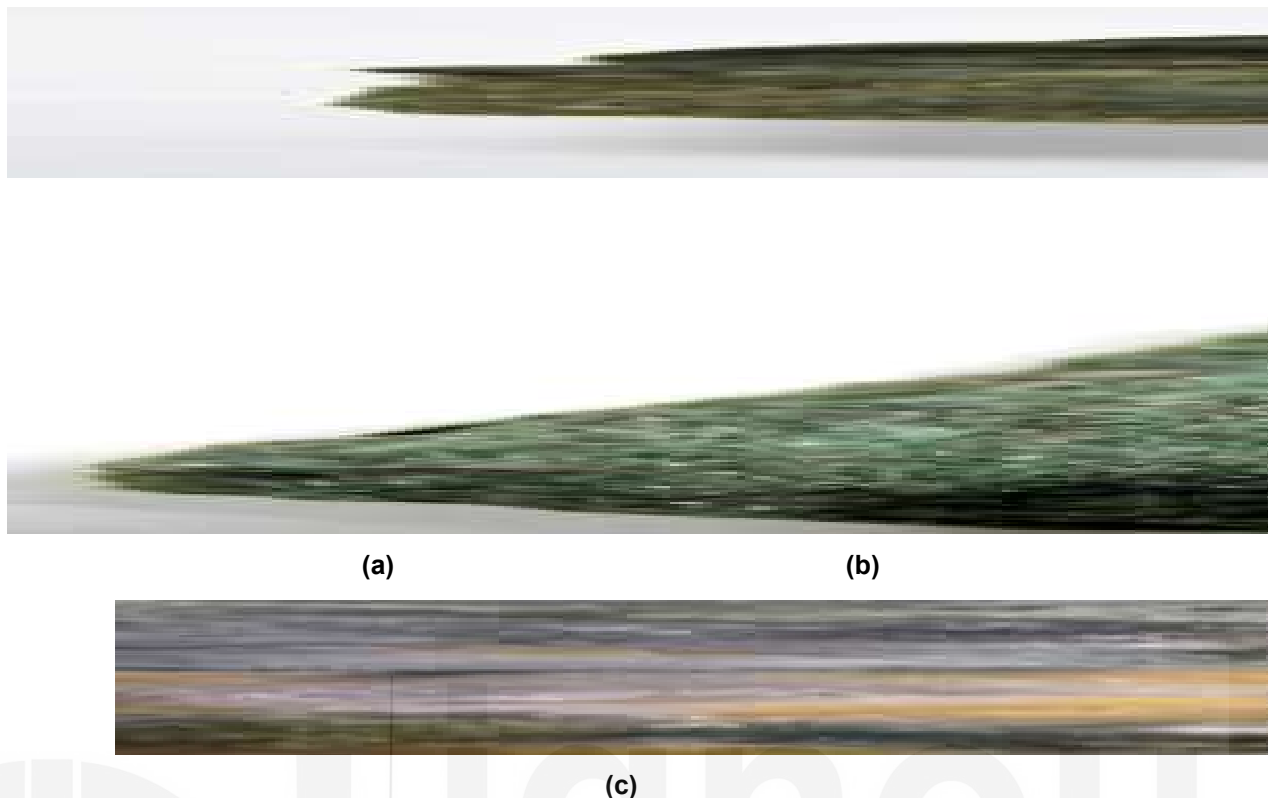


Fig. 9.4: Photographs of copper ores: a) Chalcopyrite; b) Malachite; and c) Cuprite (copper red) associated with chalcopyrite (golden yellow).

Table 9.9: Physical properties of cuprite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	Cu_2O (Cu: 88.8%)
2.	Crystal system	Cubic/Isometric
3.	Form/Habit	Massive (Fig. 9.3d), earthy
4.	Colour	Copper-red
5.	Cleavage	Poor
6.	Fracture	Uneven, brittle, conchoidal
7.	Hardness	3.5 – 4
8.	Specific gravity	5.8 - 6.1
9.	Lustre	Adamantine, submetallic, earthy
10.	Streak	Dark brownish red with shine
11.	Diagnostic properties	Colour - Copper-red Lustre - Adamantine, submetallic, earthy Streak - Dark brownish red with shine

9.4.4 Lead Ores

Owing to strong geochemical affinity, lead (Pb) is usually, associated with zinc (Zn). Lead is largely used in the manufacture of batteries. **Galena is chief ore of the lead and is described as galena.** You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.10 for galena (Fig. 9.5).

Table 9.10: Physical properties of galena.

S.N.	Physical properties	Characteristics
1.	Chemical composition	PbS
2.	Crystal system	Cubic/Isometric
3.	Form/Habit	Massive, granular, cubic (Fig. 9.5)
4.	Colour	Lead-grey to steel-grey
5.	Cleavage	Perfect cubic
6.	Fracture	Brittle, flat, even
7.	Hardness	2 – 3
8.	Specific gravity	7.4 - 7.6 (very high)
9.	Lustre	Metallic
10.	Streak	Lead grey
11.	Diagnostic properties	Colour - Lead gray Cleavage - Cubic Hardness - Low Specific gravity - Very high Streak - Lead grey with shine



Fig. 9.5: Hand specimens of galena.

9.4.5 Zinc ore

As mentioned earlier, zinc commonly occurs in association with the lead.

Sphalerite is one of the chief ore of zinc and we will study about it. You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.11 for sphalerite (Fig. 9.6).

Table 9.11: Physical properties of sphalerite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	ZnS
2.	Crystal system	Cubic, tetrahedral
3.	Form/Habit	Tetrahedra and rhombdodecahedra, often twinned and occurs as massive, compact (Fig. 9.6a)
4.	Colour	Black or brown sometimes yellow or white, rarely colourless.

5.	Cleavage	Perfect parallel to the rhombic dodecahedron faces
6.	Fracture	Brittle, conchoidal
7.	Hardness	3.5 - 4
8.	Specific gravity	3.9 - 4.2
9.	Lustre	Resinous, Adamantine
10.	Streak	White to pale yellow to off white to reddish brown
11.	Diagnostic properties	Cleavage - Perfect



Fig. 9.6: Hand specimen of sphalerite mineral.

9.4.6 Aluminium Ores

Aluminium is most widespread metal on the earth, making up more than 8% of the Earth's Core mass. It easily binds with other elements; hence pure aluminium does not occur in nature. **The most common ore of aluminium is bauxite.** You will identify their physical properties and write the chemical composition, crystal system and diagnostic properties as given in Table 9.12 for bauxite (Fig. 9.7a and b).

Table 9.12: Physical properties of bauxite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{Al}_2\text{O}_3 \cdot n\text{H}_2\text{O}$ Bauxite is a mixture of diaspora, boehmite, and gibbsite in different amounts
2.	Crystal system	Monoclinic, mostly occurs as amorphous in earthy granular or pisolitic masses, rarely crystalline

3.	Form/Habit	Pisolitic structure (Fig 9.7a), granular (Fig 9.7b), amorphous
4.	Colour	Dirty white, greyish, yellow or reddish brown
5.	Cleavage	Absent
6.	Fracture	Earthy
7.	Hardness	1 - 3
8.	Specific gravity	2 - 2.5 (Low)
9.	Lustre	Dull, earthy
10.	Streak	White
11.	Diagnostic properties	Form – Pisolitic structure, granular Colour - Yellowish, Reddish brown Hardness - 1 – 3 : Streak - White



(a)



(b)

Fig. 9.7: Hand specimens of bauxite: a) Pisolitic; and b) Granular or Oolitic form.

9.5 PHYSICAL PROPERTIES OF NON-METALLIC MINERALS

Let us examine the megascopic properties of some common non-metallic (industrial) minerals such as muscovite, gypsum, magnesite and asbestos.

9.5.1 Mica

Muscovite or white mica, and phlogopite, or amber mica, are the principal commercial micas. White mica is associated with the pegmatite and granite intrusive. Large crystals or books separable into very thin sheets, bend without breaking and snap back when bent and released are considered highly commercial. The sheets must be free from checks and crinkles. Colours or impurities of hematite and other dark minerals are found that lower the value. Muscovite mined in India is usually wine-colored or purplish. Here you will identify muscovite (Table 9.13 and Fig. 9.8a) and phlogopite (Table 9.14 and Fig. 9.8b) type of micas.

Table 9.13: Physical properties of muscovite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{H}_2\text{KAl}_3(\text{SiO}_4)_3$
2.	Crystal system	Monoclinic
3.	Form/Habit	Six sided hexagonal plates, disseminated scales or massive (Fig. 9.8a)
4.	Colour	Thick specimen often appear black, brown, or silver; thin sheet is colorless, white, sometimes with a tint of brown, yellow, green or rose
5.	Transparency	Transparent
6.	Cleavage	Perfect
7.	Fracture	Conchoidal to uneven
8.	Hardness	2.5 to 3
9.	Specific gravity	2.77 to 2.88
10.	Lustre	Pearly to vitreous
11.	Streak	White, often separable in to tiny flakes
12.	Diagnostic properties	Colour – Thin sheets as usually colourless or white or pale yellow Cleavage - Perfect Transparency - Transparent



(a)



(b)

Fig. 9.8: Hand specimens of: a) Muscovite; and b) Phlogopite.

Table 9.14 Physical properties of phlogopite.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{K}_2(\text{Mg, Fe})_6[\text{Si}_6\text{Al}_2\text{O}_{20}](\text{OH, F})_4$ (Mg:Fe is >2:1)
2.	Crystal system	Monoclinic
3.	Form/Habit	Six sided, tabular, large plates, massive and disseminated in scales (Fig. 9.8b)

4.	Colour	White, colourless, black, brown, yellow or green.
5.	Transparency	Transparent to translucent
6.	Cleavage	Perfect, parallel to basal pinacoid
7.	Fracture	Conchoidal to uneven
8.	Hardness	2 - 2.5
9.	Specific gravity	2.76 - 2.9 (low)
10.	Lustre	Pearly
11.	Streak	White, often separable in to tiny flakes
12.	Diagnostic properties	Form - Crystalline, scales Colour - White, colourless, brown, copper red Lustre - Pearly

9.5.2 Gypsum

Temporally, gypsum is associated with the sedimentary rocks of almost all the ages of shale, sandstone, and limestone. It is often found with rock salt. Almost valueless mineral, anhydrite (CaSO_4) is found with gypsum, and may be mistaken for it; but it is harder, heavier, and breaks into square pieces. Please refer Table 9.15 and Fig. 9.9 to identify gypsum mineral.

Table 9.15: Physical properties of gypsum.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
2.	Crystal system	Monoclinic (Gypsum type)
3.	Form/Habit	Crystalline (Fig. 9.9), combination of prism and clinopinacoid, twinned crystals are of (a) swallow –tail and (b) arrow-head types
4.	Colour	Colourless to white, may be yellow, tan, blue, pink, brown, reddish brown or grey due to impurities
5.	Transparency	Translucent
6.	Cleavage	Perfect, parallel to clinopinacoid
7.	Fracture	Subconchoidal
8.	Hardness	1.5 - 2
9.	Specific gravity	2.31 to 2.33
10.	Lustre	Vitreous to silky, pearly
11.	Streak	White
12.	Diagnostic properties	Cleavage - Perfect Form – Crystalline, massive Hardness – Low



Fig. 9.9: Hand specimens of gypsum.

9.5.3 Magnesite

Magnesite, commonly results from the alteration of magnesium-rich rocks during low grade metamorphism while they are in contact with carbonate-rich solutions. Magnesite has the same crystal structure of calcite, hence its inclusion into the calcite mineral group. Many of the properties of magnesite are either identical or similar to those of calcite. However, the magnesium ion does not allow the carbonate ion (CO_3) to interact as easily with cold acids, as the calcium ion does in calcite. This provides the best means of distinguishing magnesite from calcite. Please refer Table 9.16 and Fig. 9.10 to identify magnesite mineral.

Table 9.16: Physical properties of magnesite

S.N.	Physical properties	Characteristics
1.	Chemical composition	MgCO_3
2.	Crystal system	Hexagonal-trigonal (calcite type)
3.	Form/Habit	Massive (Fig. 9.10), fibrous, compact, granular, rarely crystalline
4.	Colour	White or grey, also tinted yellow or brown
5.	Cleavage	Perfect, rhombohedral
6.	Fracture	Flat or conchoidal
7.	Hardness	3.5 - 4.5
8.	Specific gravity	3 - 4.8
9.	Lustre	Vitreous or pearly, earthy, dull in massive or fibrous varieties
10.	Streak	White to off-white
11.	Diagnostic properties	Form/Habit - Rhombohedral with curved faces, massive, fibrous Cleavage - Perfect, in three directions forming rhombohedrons Gives effervescences when treated with hot acids



Fig. 9.15: Hand specimen of magnesite.

9.5.4 Asbestos

Asbestos includes fibrous forms of amphibole. The fibers in general are very long, thin and flexible. In strictest sense, the term asbestos is confined to the fibrous form of actinolite but common asbestos include fibrous varieties of a number of different silicates. The following minerals are now included in commercial asbestos:

- Chrysotile
- Actinolite
- Amosite
- Crocidolite

Please refer Table 9.17 and Fig. 9.11 to identify asbestos.

Table 9.17: Physical properties of Asbestos.

S.N.	Physical properties	Characteristics
1.	Chemical composition	$\text{Ca}_2(\text{Mg,Fe})_5[\text{Si}_8\text{O}_{22}](\text{OH,F})_2$
2.	Crystal system	Monoclinic
3.	Form/Habit	Long bladed prismatic crystals, fibrous (Fig. 9.11), massive
4.	Colour	White to greenish
5.	Cleavage	Perfect
6.	Fracture	Uneven
7.	Hardness	5 – 6
8.	Specific gravity	2.9 – 3.4 (low)
9.	Lustre	Vitreous or silky
10.	Streak	White
11.	Diagnostic properties	Colour-White to greenish Form – Massive, fibrous Lustre - Vitreous or gray Streak - White



Fig. 9.11: White fibrous asbestos.

9.6 LABORATORY EXERCISES

You have studied chemical composition, crystal system and physical properties of some of the common ores and minerals (iron, manganese, copper, lead, zinc and aluminium) and non-metallic minerals (mica, gypsum, magnesite and asbestos).

While performing this experiment, you will have to perform the following activities:

1. Observe these minerals in hand specimens by naked eyes very carefully.
2. Record their physical properties on your laboratory file, and
3. Identify the given specimens.

You should keep in your mind that in any case, no harm should be done to the hand specimens while undertaking their megascopic study.

We are listing below a performa that should be used for recording the physical properties of all the given specimens provided to you. You will record all the exercises as Laboratory Exercise 1, Exercise 2, Exercise 3 and so on. By following above mentioned procedure. You will have to describe all the given specimens of the minerals/ores one by one.

Date:

Exercise 1.....: **Metallic/ Non-metallic Mineral 1.....**

S.N.	Physical properties	Characteristics
1.	Chemical composition	
2.	Crystal system	
3.	Form/Habit	
4.	Colour	
5.	Cleavage	
6.	Fracture	
7.	Hardness	
8.	Specific gravity	
9.	Lustre	
10.	Streak	
11.	Diagnostic properties	
	Inference (Name of the metallic/ non- metallic mineral)	

9.7 REFERENCES

- Farndon, J. (2016) Rocks, Minerals and Gems. Hermes House Anness Publishing Ltd, London.
- Read, H.H. (Rev. ed. C.D. Gribble) (1988) Rutley's Elements of Mineralogy" (27th Edition), CBS Publications.
- Shrivastava, J.P. (2007) Introduction to Ore Microscopy, PHI Learning Pvt. Ltd.

9.8 LEARNING RESOURCES

- Hydrothermal Mineralisation
Link: <http://egyankosh.ac.in/handle/123456789/53461>
- Contact Metasomatic and Contact Metamorphic Deposits-
Link: <http://egyankosh.ac.in/handle/123456789/53458>
- Late Magmatic Deposits
Link: <http://egyankosh.ac.in/handle/123456789/53457>
- Early Magmatic Deposits
Link: <http://egyankosh.ac.in/handle/123456789/53456>
- Classifications of Ore Deposits
Link: <http://egyankosh.ac.in/handle/123456789/53455>
- Introduction to Ore Deposits
Link: <http://egyankosh.ac.in/handle/123456789/53454>
- Minerals & their uses
Link: <http://egyankosh.ac.in/handle/123456789/53487>
- <https://study.com/academy/lesson/what-is-ore-definition-types-uses-examples.html>
- https://www.youtube.com/watch?v=32NG9aeZ7_c
- <https://study.com/academy/lesson/what-is-pyrite-fools-gold-properties-definition-facts.html>